

Steering capabilities of an acoustic lens for transcranial therapy: numerical and experimental studies

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Abstract— For successful brain therapy, transcranial focused ultrasound must compensate for the time shifts induced locally by the skull. The patient-specific phase profile is currently generated by multi-element arrays which, over time, have tended towards increasing element count. We recently introduced a new approach, consisting of a single-element transducer coupled to an acoustic lens of controlled thickness. By adjusting the local thickness of the lens, we were able to induce phase differences which compensated those induced by the skull. Nevertheless, such an approach suffers from an apparent limitation: the lens is *a priori* designed for one specific target. In this paper, we demonstrate the possibility of taking advantage of the isoplanatic angle of the aberrating skull in order to steer the focus by mechanically moving the transducer/acoustic lens pair around its initial focusing position. This study, conducted on three human skull samples, demonstrates that tilting of the transducer with the lens restores a single -3dB focal volume at 914 kHz for a steering up to ± 11 mm in the transverse direction, and ± 10 mm in the longitudinal direction, around the initial focal region.

Acoustic lens, transcranial ultrasound, therapeutic ultrasound

I. INTRODUCTION

Focused ultrasound has been applied to a growing number of pathologies: prostate [1, 2] or pancreatic [3, 4] cancers, bone metastases [5, 6], uterine fibroids [7-9], or essential tremor [10-12]. It complements or even advantageously replaces surgery-based conventional methods. One of the prime advantages of the method is that it is non-invasive or minimally-invasive, and thus reduces convalescence time and post-operative risks of complications. It has however remained inapplicable to the human brain for a long time, due to the defocusing effect of the skull [13, 14]. In the 70s, transcranial aberration correction techniques were introduced in the context of human brain imaging [15, 16]. Then, in the 90s, wavefront control was demonstrated in vitro for transcranial therapeutic ultrasound

[17-21], before being tested on animals [22-24] and cadavers [25-27]. Transcranial focusing can be used with a diverse range of *modi operandi*, including Blood-Brain-Barrier (BBB) opening [28-32], thermal lesioning [23, 24, 33], and neuromodulation [34-40], some of which are already being used in clinics [10, 41-43].

The phase-corrected profile is usually generated by a multi-element array: each element emits a signal with a precomputed phase so as to compensate for the time shifts induced locally by the skull. For a fixed dimension of the element array, the quality of the focusing being directly linked to the number of elements, the progress made in electronics and transducer conception led to the development of probes featuring an increasing number of elements: 64 elements in 2000 [44], 200 in 2003 [45], 512 in 2013 [25], 1024 in 2010 [46] or 1372 also in 2010 [47]. Inspired by previous studies employing acoustic lenses for therapeutic ultrasound [48-55], we introduced a new approach [56], consisting of a single-element transducer coupled to an acoustic lens of controlled thickness. By adjusting the local thickness of the lens, we were able to induce phase differences which compensated those induced by the skull. With such acoustic lenses, we were able to focus through a set of 3 human skulls, and show comparable levels of precision and efficiency to those attained by conventional, multi-element systems. Nevertheless, such an approach suffers from an inherent limitation: each lens is designed for one given patient and one given target [57], meaning that – for a given patient – a surgeon could neither change the target after the fabrication of the lens, nor treat a large area over the course of the surgery, without making a new lens. Correcting aberrations with an acoustic lens thus seemed like a less flexible method than using multi-element arrays, with which new phase sets can be electronically changed to aim at another target during treatment procedure [45]. In this paper, we propose to take advantage of the isoplanatic angle of the aberrating skull in order to steer the focus by mechanically moving the transducer/acoustic lens pair around its initial

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focusing position, similarly to what is done in astronomy to compensate the aberrations induced by the atmosphere [58]. The isoplanatic angle is the angle between two targets for which the heterogenous medium crossed by the wave is similar. In such conditions, the aberration corrections computed for the first target remains efficient for the second one. This study was conducted on three human skulls, first numerically, and then experimentally. The quality of the focusing is assessed for a longitudinal and transverse translation of the focus.

II. MATERIALS AND METHODS

A. Transducer

A single-element transducer with a 61mm radius of curvature and a 67mm aperture (H101 MR, Sonic Concepts, Bothell, USA) was used to achieve transcranial focusing. It was operated at its resonance frequency (914kHz), which was determined from the analysis of its complex impedance against frequency, using a vector network analyzer (ZVL, Rohde & Schwarz, Munich, Germany). Prior to the study, we evaluated

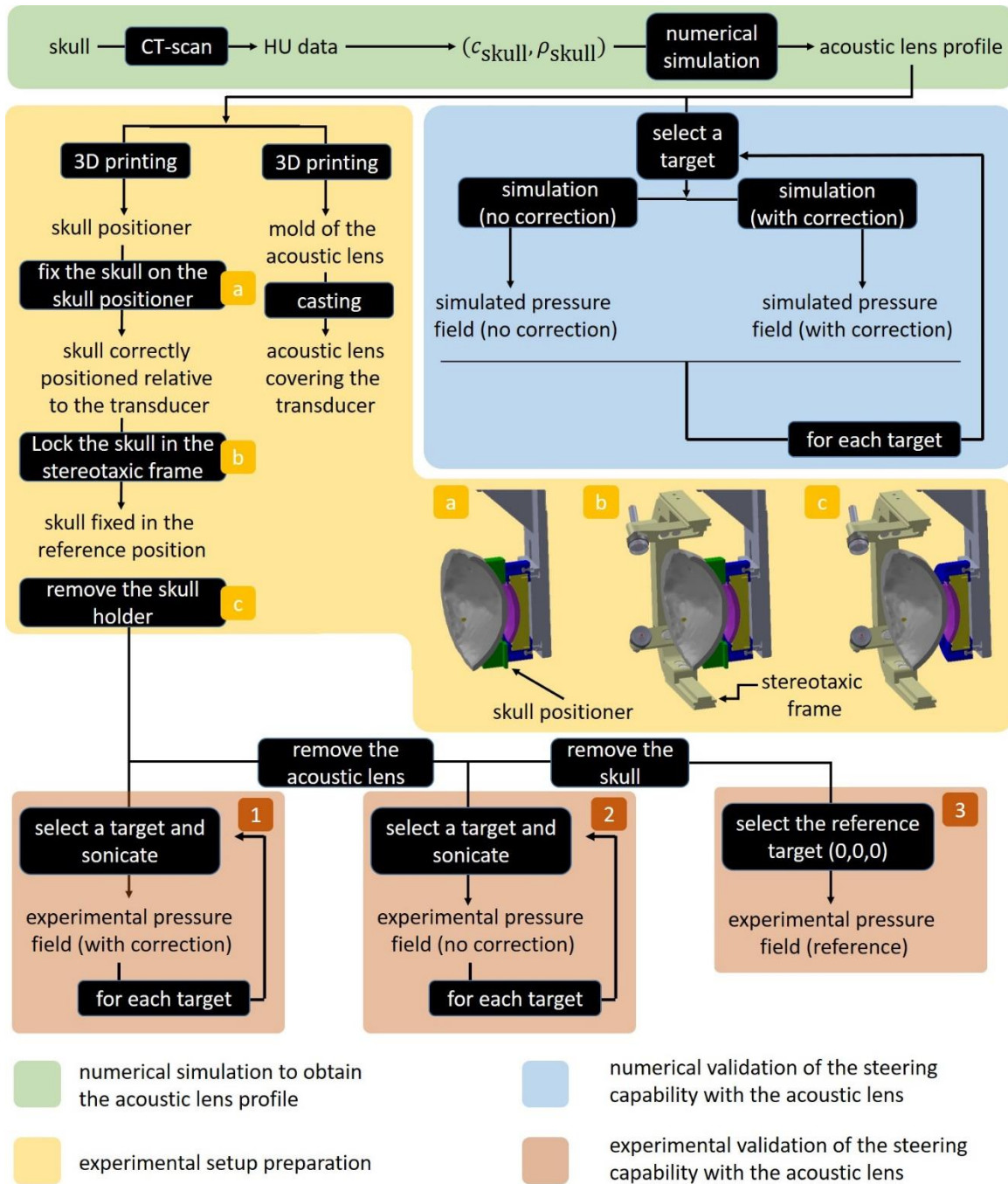


Figure 1: Synoptic view of the whole numerical and experimental protocols. The four major phases are colored for easier identification.

that this transducer had a 91% electro-acoustic conversion efficiency at its resonance frequency.

B. Human skull

The experiments were conducted on $N = 3$ human skulls (hereafter referenced as A, B and C). The human skulls were provided by the Institut d'Anatomie (UFR Biomédicale des Saints-Pères, Université Paris Descartes, Paris, France) and tattooed with individual numbers, as approved by the ethics committee of the Centre du Don des Corps (Université Paris Descartes, Paris, France). The skulls were immersed in water and degassed under a 2mbar reduced pressure (diaphragm pump FB65457, Fischer Scientific, Waltham, MA, USA) for 48h; their CT-scans were acquired (Sensation 64, Siemens, Erlangen, Germany) at the CHU de Marseille Nord, France. The in-plane spatial resolution of the slices was 0.41mm, while slice thickness and interslice spacing were 0.60mm and 0.30mm, respectively. The speed of sound and density maps of the skull used in the numerical simulations were derived by a linear approximation from these HU data [59].

C. Determining the thickness of the acoustic lens through numerical simulation

In order to determine the thickness of the corrective acoustic lens, an acoustic wave was emitted from the target point (Figure 1, green box) and numerically propagated across the skull. The simulated emitter was placed at a known location relative to the skull, which enabled us to target a predefined region. A 3-dimension linear interpolation of the speed of sound and density maps was performed to obtain a $\lambda_{water}/10$ spatial step in each dimension. The time step Δt_{sim} was computed based on a 0.3 Courant–Friedrichs–Lewy (CFL) condition in the cortical bone. The simulation was conducted using a k-space pseudospectral method-based solver, k-Wave [60]. Neither the absorption properties of the skull nor those of the tissues were taken into account and the skull bone is assumed to be a fluid medium. To further reduce the simulation time, the point source at the target was replaced with an equivalent spherical wavefront originated from the target and adjusted as closely as possible to the brain-skull interface. Figure 4 shows a cross-section view of the

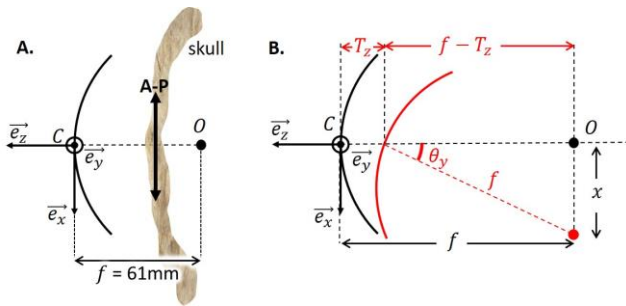
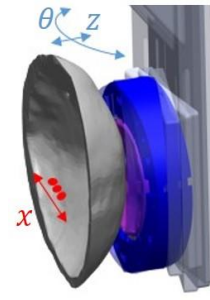
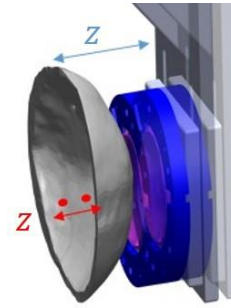


Figure 3: A. Coordinate system used in the study. Its origin O corresponds to the target, the x direction to the Antero-Posterior (A-P) axis of the skull. B. Evidencing the z and θ_y movements required to steer toward the x direction.

numerical simulation setup. A 2-cycle sinusoidal burst with Gaussian apodization was emitted at the frequency $f_0 = 914$ kHz and numerically propagated in the medium. The signal was



	x-steering (mm)	z-stage (mm)	θ_y -stage	SKULL		
				A	B	C
x_5'	-11.2	-1.0	$10^\circ 35'$	•	•	•
x_4'	-9.0	-0.7	$8^\circ 28'$	•	•	•
x_3'	-6.8	-0.4	$6^\circ 21'$	•	•	•
x_2'	-4.5	-0.2	$4^\circ 14'$	•	•	•
x_1'	-2.3	0.0	$2^\circ 7'$	•	•	•
x_0	0.0	0.0	0°	•	•	•
x_1	2.3	0.0	$2^\circ 7'$	•	•	•
x_2	4.5	0.2	$4^\circ 14'$	•	•	•
x_3	6.8	0.4	$6^\circ 21'$	•	•	•
x_4	9.0	0.7	$8^\circ 28'$	•	•	•
x_5	11.2	1.0	$10^\circ 35'$	•	•	•



	z-steering (mm)	z-stage (mm)	θ_y -stage	SKULL		
				A	B	C
x_0	0.0	0.0	0°	•	•	•
x_1	2.0	2.0	0°	•	•	•
x_2	4.0	4.0	0°	•	•	•
x_3	6.0	6.0	0°	•	•	•
x_4	8.0	8.0	0°	•	•	•
x_5	10.0	10.0	0°	•	•	•

Figure 2: Longitudinal and transverse steering configurations. The values of the z translation and of the θ_y rotation are provided for each.

recorded on the surface of the transducer and the phase φ of the signal was evaluated by a projection onto the Fourier basis. The lens thickness was then 2D-unwrapped on the surface of the transducer to avoid phase discontinuities. The local thickness p of the lens on every point M of the transducer is evaluated, according to the following formula, and where $\tilde{\varphi}$ is the phase after unwrapping:

$$p(M) = \frac{\tilde{\varphi}(M)}{2\pi f_0} \cdot \frac{1}{\frac{1}{c_{\text{water}}} - \frac{1}{c_{\text{lens}}}} + K \quad (1)$$

K is a constant length chosen to set the minimal thickness of the lens. In our study, K was set to 2 mm. The speed of sound was $c_{\text{water}} = 1485$ m/s [61] and $c_{\text{lens}} = 1000$ m/s [56] respectively in water and in the acoustic lens.

The simulation was run on a workstation equipped with two 2.20GHz Intel Xeon CPU E5-2630v4 processors and 32GB RAM; the full running time was about one hour.

More details about the method used to build the acoustic lens can be found in Maimbourg *et al.* [56].

D. Mechanism of the experimental setup

A mold was 3D-printed (Replicator 2X, Makerbot, Brooklyn, NY, USA) based on the shape that had been determined from the numerical simulation (Figure 1, yellow box). The acoustic lens was then cast with a commercial silicone (Elite double 8, Zhermack, Badina Polesine, Italy). A plastic holder of shape complementary to the skull was also 3D-printed, in order to position the transducer relative to the skull in a low lost and effective manner. The positioning of the skull was achieved by a three-step process summarized in Figure 1, yellow box:

- The skull is positioned relative to the transducer thanks to the skull positioner.
- The screws of the stereotactic frame are tightened so that the skull remains still.
- The transducer is moved backwards, and the skull positioner removed.

At the end of these steps, the skull was in a known position relative to the transducer, all the while allowing for the movement of the transducer relative to the skull, which allowed us to test mechanical steering. Figure 2.A indicates the geometrical coordinate system chosen for the study. The transducer was then mounted onto a linear stage (UE71PP, Micro-Controle, Newport, CA, USA) on the z axis and a rotary stage (TR80-BL, Micro-Controle, Newport, CA, USA) on the (Cy) axis, where C is the point corresponding to the center of the surface of the transducer. Using this setup, the precision was 0.01mm in linear A-P displacement, and 1° in rotation. These two degrees of freedom allowed for steering along the x (anterior-posterior axis of the skull) and z (superior-inferior axis of the skull) axes. Figure 3, as well as the corresponding table, recapitulate the targeted positions, along with the θ_y rotations and T_z translations required to reach them. To obtain an x -axis translation, we combined a θ_y rotation and a T_z translation (Figure 2.B). With $f = 61$ mm being the focal distance of the transducer, the following formulae link θ_y and T_z to the desired x translation:

$$\theta_y = \sin^{-1}\left(\frac{x}{f}\right) \quad \text{and} \quad T_z = f - \sqrt{f^2 - x^2} \quad (2)$$

Due to the proximity between the acoustic lens and the skull, every steering configuration was not feasible. Indeed, for certain configurations, the acoustic lens interfered with the skull. It was not possible to attempt steering in the $+\vec{e}_z$ direction in particular, and skull C was the only one that could allow testing for positive and negative steering along x . We assumed that steering positions with positive and negative x were equivalent for the purpose of the study. As a consequence, the

positive and negative steerings for skull C were concatenated in the Results section. The tables in Figure 3 show the tested configurations for each skull. During a preliminary study, we used a first transducer with a different focal depth for which a 2mm displacement along x was achieved by incrementing θ_y by $2^\circ 7'$. For practical reasons, we kept this angular step throughout the rest of our study, resulting in the x -steering step being a non-round value (for $\theta_y = 2^\circ 7'$, the x -steering equals 2.3mm).

E. Numerical assessment of the steering capability

We first numerically assessed the capabilities of the method (Figure 1, blue box). To this end, a sinusoidal wave was emitted at 914 kHz from the surface of the transducer and propagated through the acoustic lens and the skull (Figure 4, right). The numerical simulation was conducted according to the same specifications (speed of sound and density maps, spatial step, CFL conditions) used when determining the thickness of the lens. It was, however, conducted on a larger volume that included the target (white dashed box, on Figure 4, right). The pressure field was recorded on a $x \times y \times z = 8 \times 8 \times 16$ mm³ volume around the target. In order to simulate focusing at

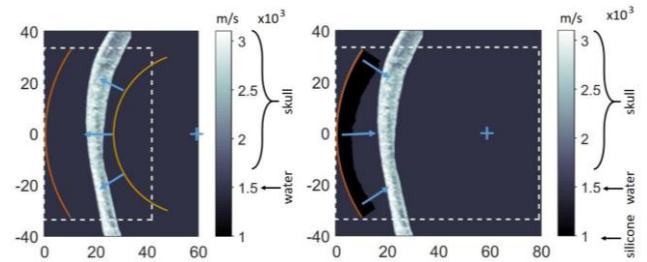


Figure 4: Left: Speed of sound map of skull A in a cross-section plane defined by $x=0$. The blue cross represents the target and coincides with the geometrical focus of the transducer. The yellow arc centered on the target is the source used in the simulation. The red surface represents the surface of the transducer where the acoustic signal is recorded, after propagation through the skull. The white dash box delimits the area in which the simulation was calculated. Right: The wave is emitted from the surface of the transducer and propagates through the acoustic lens and the skull. The pressure field is acquired in a volume centered on the target.

different positions, the speed of sound and density maps representing the skull were rotated and displaced while the transducer and the lens remained static. All the positions that were experimentally tested (Figure 3) were tested numerically first.

Moreover, to quantify the effect of the ultrasound frequency on the acoustic lens correction capabilities and on the influence of the steering, we have also performed this numerical evaluation for two other frequencies: 220kHz and 660kHz. These two frequencies correspond to those used on the commercial brain therapy system developed by Insightec. We restricted the study to steering along x for the positions 2.3mm, 6.8mm and 11.2mm.

F. Experimental assessment of the steering capability

The pressure field was recorded for each steering position (Figure 1, red boxes). An arbitrary waveform generator (AFG3101C, Tektronix, Beaverton, OR, USA) emitted a 200-cycle sinusoidal burst at 914kHz. This signal was then amplified (40AD1, Amplifier Research, Souderton, PA, USA). An oscilloscope (DPO3034, Tektronix, Beaverton, OR, USA)

Onda Corp., Sunnyvale, CA, USA) before digitalization at the sampling frequency $f_s = 100\text{MHz}$ with $N = 1\text{MPts}$ using a data acquisition card (HS5, TiePie, Sneek, Netherlands). Prior to the experiments, the hydrophone was calibrated according to an absolute measure of pressure, using a heterodyne interferometer [62]. A 1.40V/MPa sensitivity was obtained. All the devices were synchronized using a computer, thanks to an

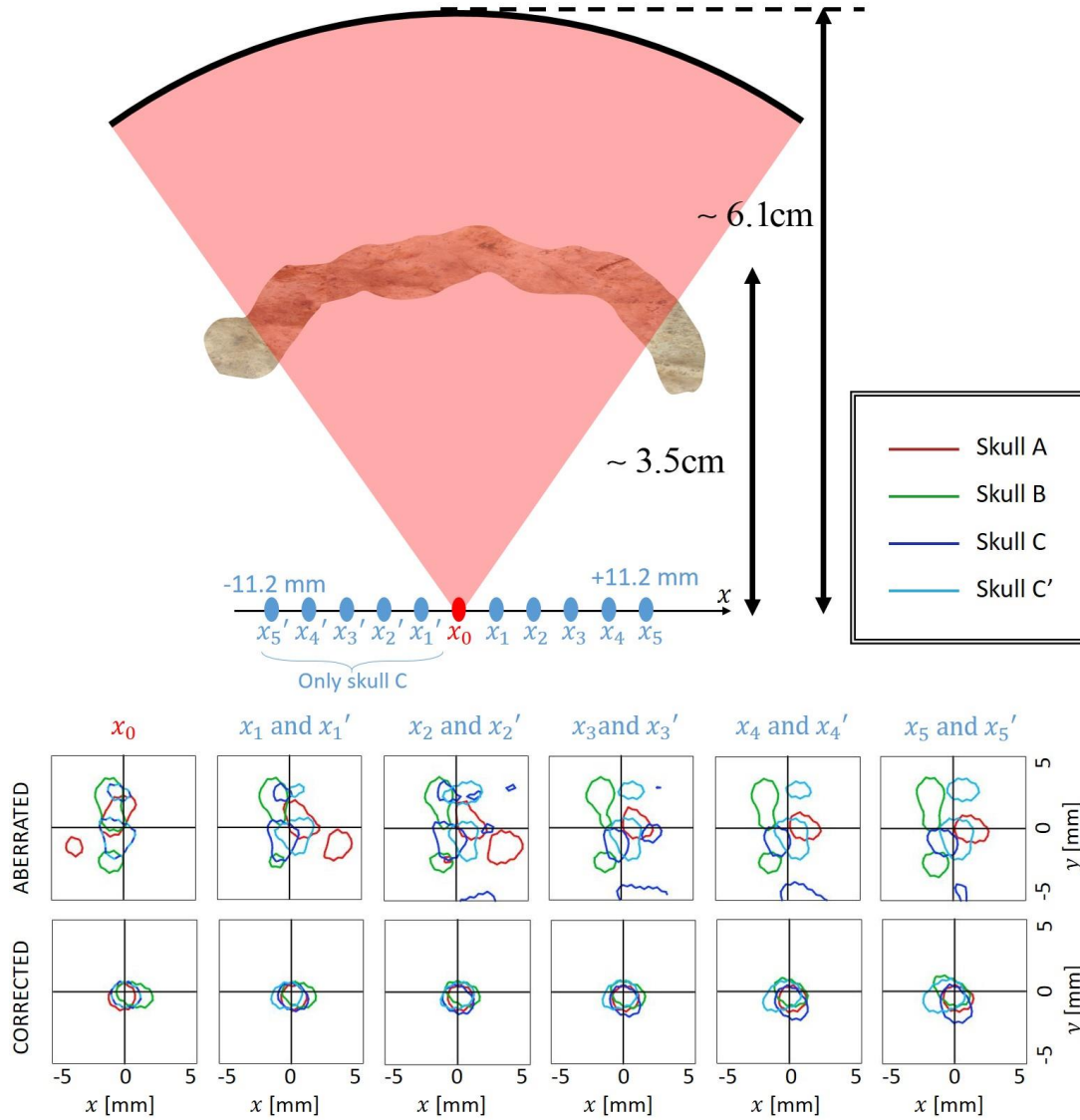


Figure 5: -3dB focus area in the transverse plane for a transducer mechanically steered along the x axis. The results for the three skulls (and for both positive and negative steering, in the case of skull C) are presented on top of one another. The top row displays the results obtained experimentally for a case where the wave propagated across a skull without the correction of the acoustic lens. The bottom row displays the results obtained after correction by the acoustic lens. The black cursor represents the desired target.

recorded the input and output signals of the amplifier. The acoustic pressure field was measured with a needle hydrophone (HNA-0400, Onda Corp., Sunnyvale, CA, USA) mounted on a 3-axis positioning system (ESP301 and ILS Linear Stages, Newport, CA, USA). The x , y and z axes corresponded to the linear stages, so that z corresponded to the longitudinal axis of the transducer (Figure 2.A). The output signal of the hydrophone was amplified (AG-2020 and AH-2020-DCBSW,

in-house Matlab code (Mathworks, Natick, MA, USA). For each position of the hydrophone, the signal was averaged throughout 5 acquisitions. In order to extract the harmonic component from the acoustic signal, the average signal was projected onto the Fourier basis corresponding to the excitation frequency (914 kHz). As illustrated in Figure 1, red boxes, the pressure field was recorded in three cases: (1) propagation through the skull with the acoustic lens correction, (2) propagation through the skull (3) propagation through water

only (only recorded for the target (0,0,0)). The pressure field obtained through water only served as a reference for the target before steering (0,0,0). Then, for example, the target for a 2mm x-steering was assumed to be located at (2,0,0). The pressure was recorded along two paths: a circular XZ plane and a circular XY plane with 6mm radius (0.3mm resolution).

G. Data analysis

The acoustic intensity I_{ac} was evaluated for each case, using the pressure fields obtained both experimentally and by numerical simulation:

$$I_{ac} = \frac{A_m^2}{2Z} \quad (3)$$

where A_m is the pressure amplitude and Z is the acoustic impedance of water.

During the experiments, the acoustic intensity obtained for the acoustic lens correction was corrected from the attenuation induced by the lens, as explained in [56]. This allows direct comparison with the simulations, in which the attenuation of the lens is not modeled. Comparisons were thus carried out with equal intensity on the outer surface of the skull.

The position of the acoustic focus was estimated by calculating the barycentric coordinates of the intensity within the -6dB main focal region.

$$\vec{OF} = \begin{pmatrix} x_F \\ y_F \\ z_F \end{pmatrix} = \frac{\sum_{V@-6dB} I_{ac}(M) \vec{OM} \Delta x \Delta y \Delta z}{\sum_{V@-6dB} I_{ac}(M) \Delta x \Delta y \Delta z} \quad (4)$$

The longitudinal error ϵ_{longi} and transverse error ϵ_{trans} were then calculated:

$$\epsilon_{trans} = \sqrt{(x_F - x_T)^2 + (y_F - y_T)^2} \text{ and } \epsilon_{longi} = |z_F - z_T| \quad (5)$$

where (x_T, y_T, z_T) is the coordinate of the target.

The surface of the -3dB focal spot on the transverse plane was finally calculated, in order to assess the dispersion of the acoustic power.

All these metrics were evaluated for skulls A, B and C, with and without the correction provided by the lens, for both experimental and numerical results.

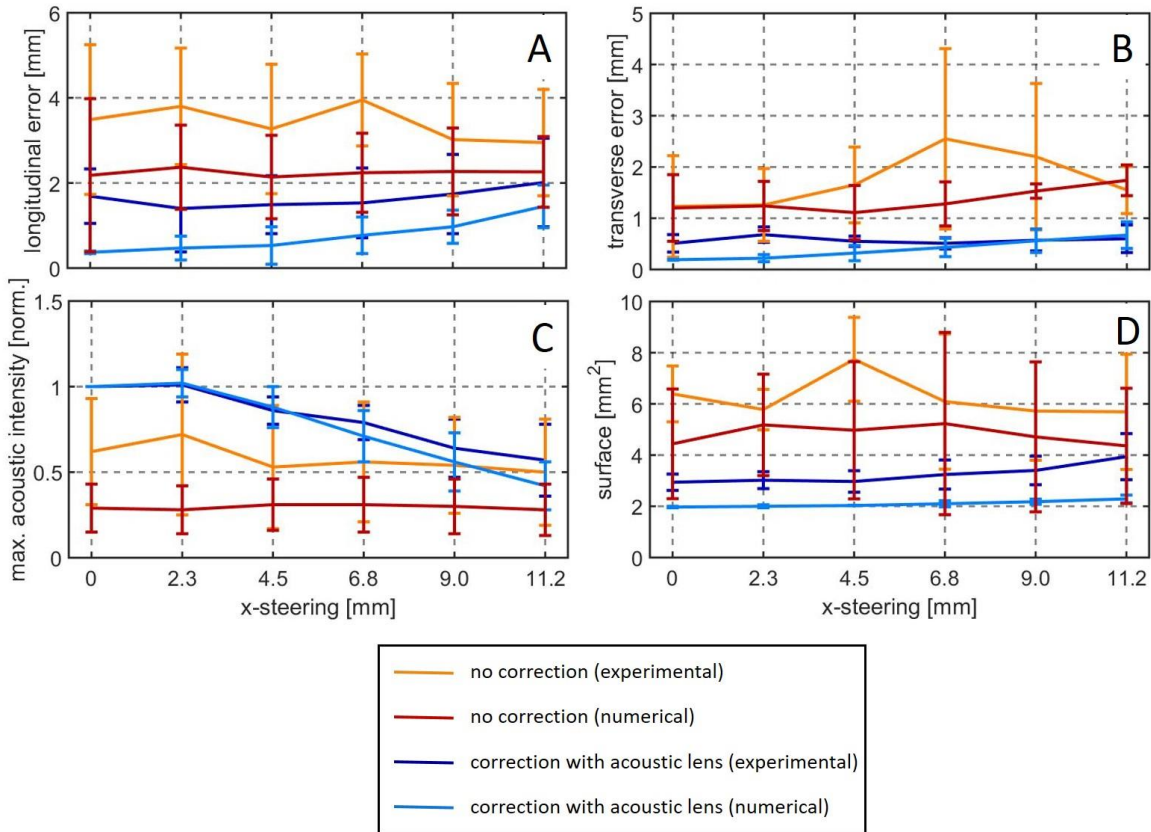


Figure 6: A) Average and standard deviation of longitudinal error ϵ_{longi} (equation 5), as a function of steering along the x axis. B) Average and standard deviation of transverse error ϵ_{trans} (equation 5), as a function of steering along the x axis. C) Average and standard deviation of maximum acoustic intensity (normalized by the acoustic intensity obtained with the acoustic lens correction without steering), as a function of steering along the x axis. D) Average and standard deviation of the -3dB area focal spot in the transverse plane, as a function of steering along the x axis.

III. RESULTS

A. Numerical and experimental evaluations of steering capabilities along the transverse direction x

Figure 5 shows the -3dB focus area for a transducer mechanically steered along the x axis. The results for the three skulls (and for both positive and negative steering, in the case of skull C) are presented on top of one another. The top row displays the results obtained experimentally without the acoustic lens correction. Without any correction, multiple foci are always present and focusing quality is poor. More energy is dispersed along the y axis (the transverse axis of the skull) than along the x axis in the majority of cases shown here. The dispersion also tends to moderately increase as the steering angle increases. The bottom row displays the results obtained with the acoustic lens correction. A unique focus was obtained in every case. The dispersion of acoustic energy does not

noticeably increase when the acoustic lens is added, and said dispersion does not noticeably increase either for x -direction steering, up to 11.2mm (positions x_5 and x_5'). The precision of the focusing suffers from greater steering, however.

Figure 6 quantitatively summarizes these results and shows the longitudinal and transverse errors, the maximum acoustic intensity, as well as the surface of the -3dB area in the XY (transverse) plane. Averages and standard deviations are calculated for all three skulls, with and without correction by the lens, for both numerical and experimental results. Without correction, the curves do not follow a clear trend relative to an increase in steering along the x axis. With the correction provided by the acoustic lens, the longitudinal and transverse errors are less than 2mm and 1mm respectively for every steering position. For an 11.2mm steering distance, the

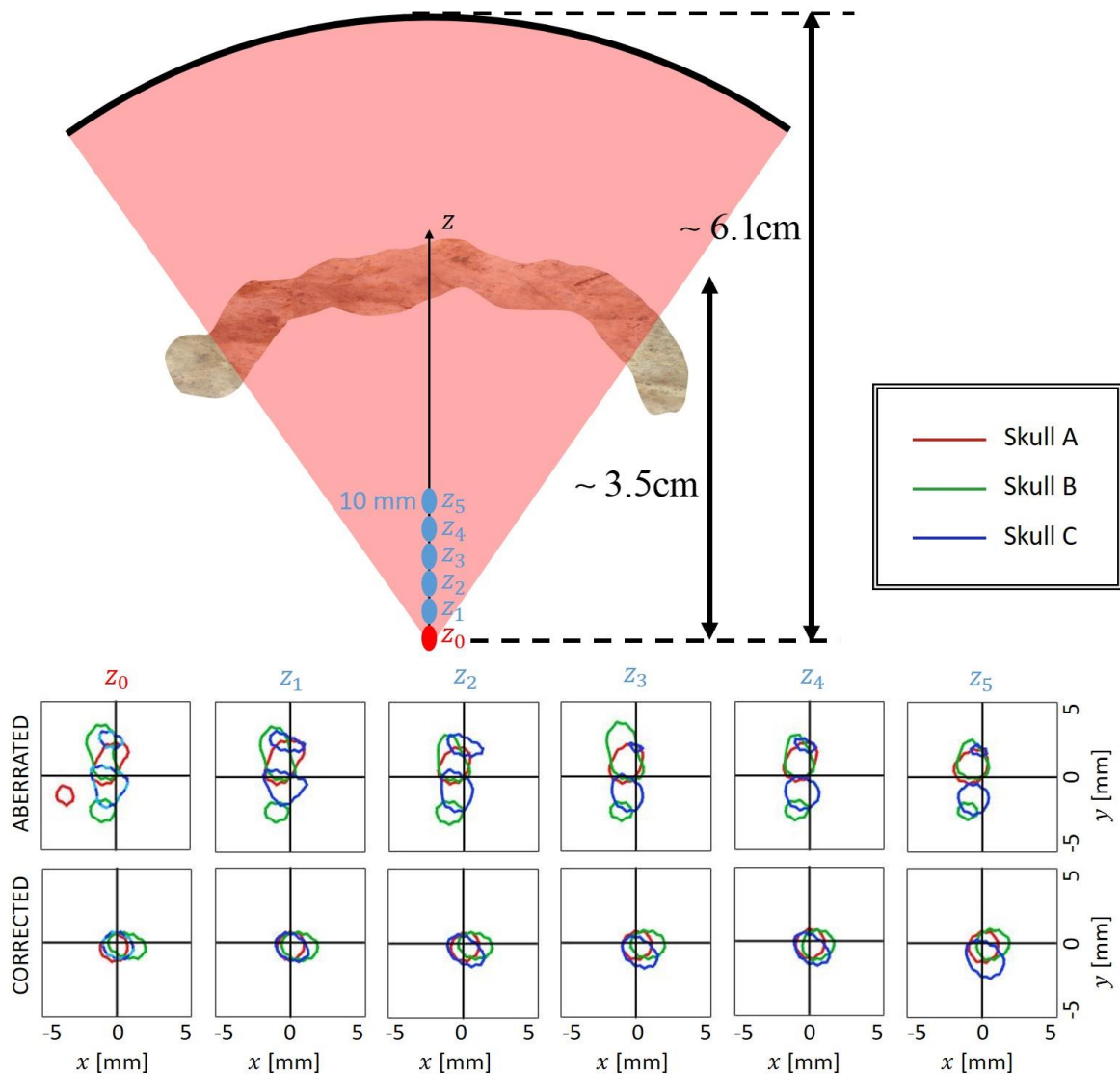


Figure 7: -3dB focus area in the transverse plane for a transducer mechanically steered along the z axis. The results for the three tested skulls are presented on top of one another. The top row displays the results obtained experimentally for a case where the wave propagated across a skull without the correction of the acoustic lens. The bottom row displays the results obtained after correction by the acoustic lens. The black cursor represents the desired target.

maximum acoustic intensity decreases by $43 \pm 21\%$ and $68 \pm 14\%$ experimentally and numerically, respectively, compared to the reference target without steering. The -3dB area in the XY plane is about half the width with correction – and increases with steering. Compared to the reference (steering-less position), it grows by $34 \pm 42\%$ experimentally and by $16 \pm 9\%$ numerically for an 11.2mm steering along the x axis.

B. Numerical and experimental evaluations of the steering capabilities along the longitudinal direction z

Figure 7 shows the -3dB focus area when the transducer is mechanically steered along the z axis. The results for the three skulls are drawn on top of one another. The top row represents experimental results through the skull without the acoustic lens correction. Without correction, the energy is dispersed across multiple foci. This dispersion does not change with z-axis steering. The acoustic lens correction restores a single focus in every case. The surface of the focal spot increases slightly with steering.

Figure 8 quantitatively evaluates the impact of z-axis steering on four significant parameters: the transverse and longitudinal errors, the maximum acoustic intensity, and the surface of the -3dB area. Without correction, these indicators remain constant and do not follow a monotonic trend when the steering depth increases. With correction, the longitudinal and transverse errors remain below 2mm and 1mm, respectively, for every steering position along the z axis. Compared to the reference (steering-less position), the acoustic intensity decreases by 37%

experimentally (and by 60% numerically) for 10mm of steering along the z axis. The -3dB area slightly increases with steering; it varies from 2.94mm^2 (no steering) to 3.72mm^2 (10mm steering) in the experimental case, and from 1.97mm^2 to 2.77mm^2 in the numerical case.

Figure 9 compares the numerical results of steering along the x axis on the maximum acoustic intensity and the transverse error, for 3 ultrasound frequencies (220kHz, 660kHz and 914kHz). The defocusing effect of the skull is weaker for lower frequencies. Indeed, without any steering, the transverse error at 220kHz is reduced from 0.62mm to 0.45mm (compared 1.2mm to 0.2mm at 914kHz) by using the acoustic lens while the maximum acoustic intensity increases by only 15% (230% at 914kHz). As expected, the benefit of aberration correction is more critical for higher frequencies. The impact of steering is similar for all the frequencies: the transverse error increases slowly and never exceeds 1mm whereas the difference in maximum acoustic intensity between no correction and lens-based correction decreases progressively and vanishes for 11.2mm steering.

IV. DISCUSSION

To put them in perspective, the positioning errors relative to the target should be compared to the focal spot dimensions through its Full Width at Half Maximum (FWHM) for free-space propagation. These amount to 11 mm in the longitudinal direction and to 1.5 mm in the transverse direction,

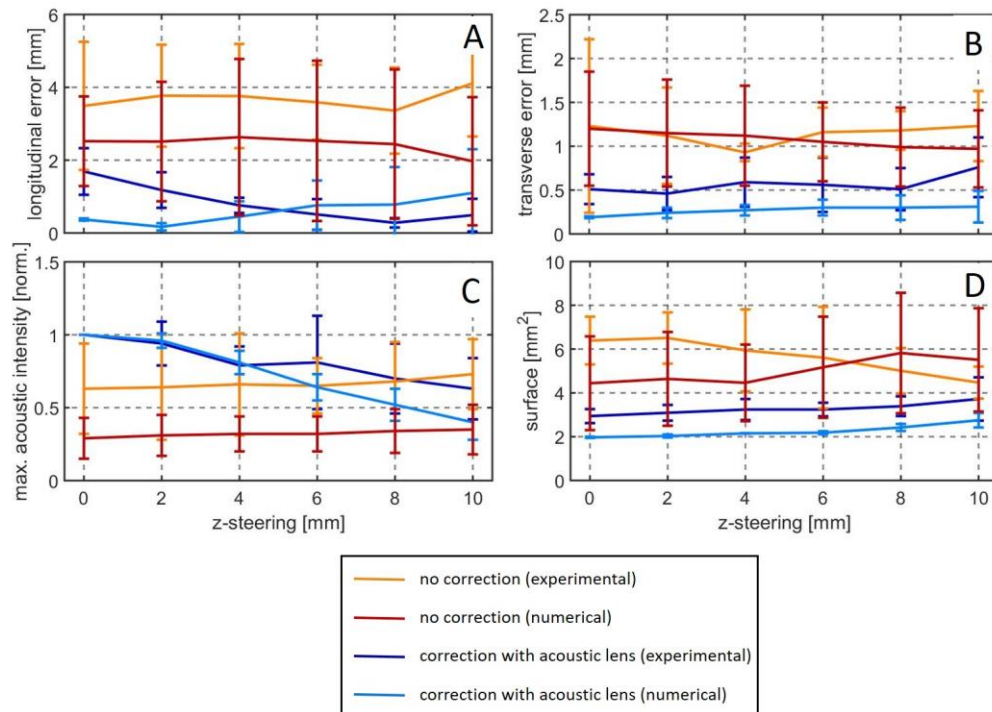


Figure 8: A) Average and standard deviation of longitudinal error ϵ_{longi} (equation 5), as a function of steering along the z axis. B) Average and standard deviation of transverse error ϵ_{trans} (equation 5), as a function of steering along the z axis. C) Average and standard deviation of maximum acoustic intensity (normalized by the acoustic intensity obtained with the acoustic lens correction without steering), as a function of steering along the z axis. D) Average and standard deviation for the surface of the -3dB focal spot in the transverse plane, as a function of steering along the z axis.

respectively. In this study, without steering, the average error is 1.69 mm in the longitudinal direction and 0.51 mm in the transverse direction. The positioning errors thus correspond to 15% and 34% of the FWHM values respectively.

This study performed at 914 kHz shows that the energy deposition significantly decreases with mechanical steering (-50% when steering 10 mm) in the longitudinal direction as well as in the transverse direction. For sake of comparison, electronic steering with a transcranial multi-elements array at a similar frequency (1 MHz) showed a 50% in energy deposition when steering 20 mm in the transverse direction [63]. As a consequence, trying to reach an identical steering range for high-intensity treatments would *in fine* require twice the acoustic power to obtain equivalent energy levels at the focus, which would mean that the patient's skull would also absorb twice as much energy. Nevertheless, compensating such steering losses would be less of an issue for low-power therapies such as BBB opening or neuromodulation.

Moreover, the steering capabilities needed for clinical translation differ with the indication. In the case of ultrasonic surgery in the treatment of essential tremors, the target is the ventral intermediate (VIM) nucleus. Unfortunately, the VIM is not visible on current MR images acquired at 1.5 T or 3 T and the neurosurgeon estimates the position of the VIM relative to the anterior-posterior commissural line [10]. From this initial target, low-power sonications are first performed. Based on the physiological reactions of the patient, the neurosurgeon adjusts the position of the focus in order to pinpoint the VIM nucleus. These intraoperative adjustments require millimeter displacements of the focus around the initial target. The steering capabilities presented in this study thus appear in line with the clinical need for essential tremor treatments. The steering

capability demonstrated here could also be applied to track a moving target through an aberrating medium by compensating in real-time the respiratory or heart motion.

In another application and for the first time, Coluccia *et al.* have performed in 2014 the partial ablation by focused ultrasound of a glioblastoma multiforme (GBM) [64]. In that specific case, the tumor volume is relatively small (6.5 cm³). In such small tumor cases, our steering capabilities are thus also sufficient. However, a retrospective study shows that on a sample of 135 patients suffering from GBM, the average tumor volume before resection could reach 34 cm³ [65], indicating tumors of width exceeding 3.5 cm. To treat those tumors or brain metastasis, BBB opening by focused ultrasound is promising. However, the 10 mm displacements of the focus obtained in this study are insufficient if one hopes to achieve focusing by steering on a volume large enough to perform a large opening of the BBB. In that case, a specific lens should be designed to target a wider volume at lower intensity.

Several sources of error contribute to the discrepancies between the numerical and experimental results (as can be seen in Figure 8 for example). First, the acoustical model of the skull is the result of previous optimizations [59] and could be further refined. Second, we estimate that the repositioning error of the skull in regard to the transducer is 1mm in translation, and less than 5 degrees in rotation. Third, shear waves are not modeled in our simulations and the relative impact of the shear waves will depend on the incidence angle on the surface of the skull.

For the purpose of this study, we used the exact same acoustic lenses used in Maimbourg *et al.* [56]. It is worth mentioning that efficiency of the lenses was not affected after a 12-months storage and manual repositioning on the surface of

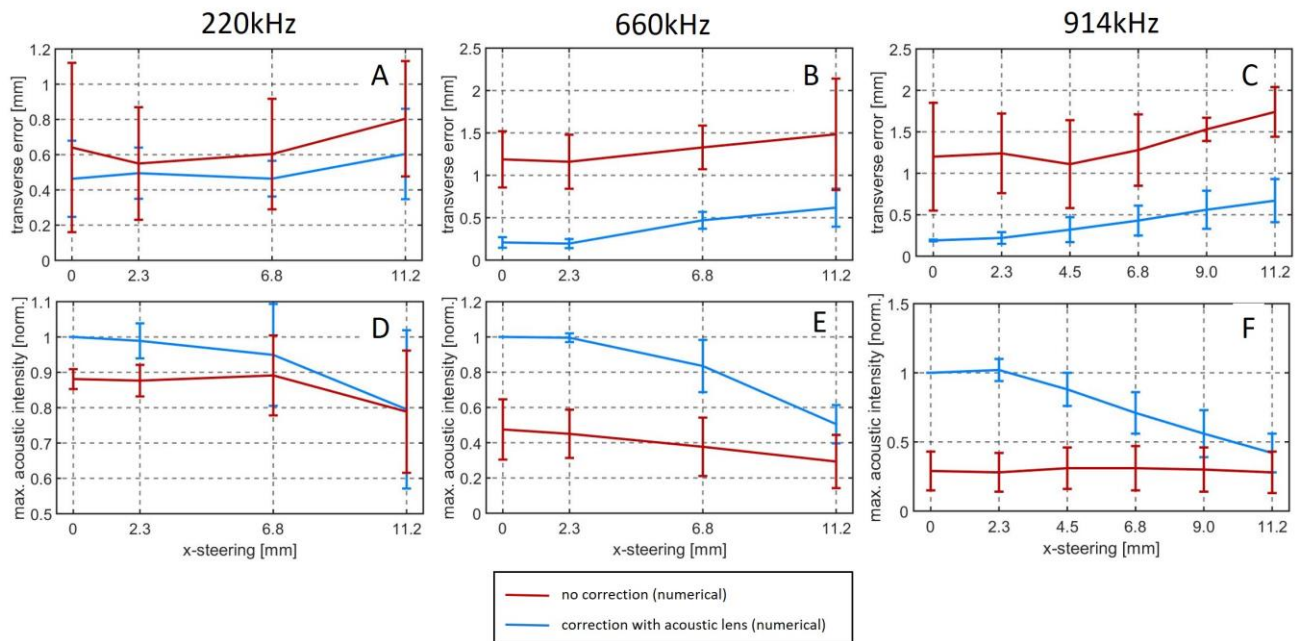


Figure 9 : A,B and C) Average and standard deviation of transverse error ε_{trans} (equation 5), as a function of steering along the x axis at 220kHz, 660 kHz and 914kHz. D, E and F) Average and standard deviation of maximum acoustic intensity (normalized by the acoustic intensity obtained with the acoustic lens correction without steering), as a function of steering along the x axis at 220kHz, 660 kHz and 914kHz.

the transducer. This highlights that the shelf-life of the lens is longer than one year.

It is worth mentioning here that the transducer used in this proof of concept paper has a high f-number ($F/D = 0.91$) compared to current transcranial therapy transducers for which F/D is ranging between 0.7 [59] and 0.5 [66-68]. Further work needs to be done with a larger aperture transducer in order to assess the potential of the lens-based correction for transcranial thermal ablation with an antenna gain similar to current clinical transcranial ultrasonic neurosurgery devices.

The lens-based correction approach is promising for a wide range of applications. We presented here its steering capabilities for transcranial focusing. The concept was applied recently to improve the focusing through cancellous bone [69], and could be used to improve cancellous bone characterization using transmission and/or backscatter measurements [70, 71].

In further developments, we would like to have a lens to control the phase and amplitude of the wave independently. To perform this, we plan to use two lenses made of materials with different acoustic properties (in sound speed and attenuation) and adjust their respective thickness to act on two degrees of freedom.

V. CONCLUSION

By contrast with a multi element array, for which the phase profile is reconfigurable in real-time, a single-element covered with an acoustic lens is designed for one given target. Nevertheless, we demonstrated here that moderate steering is achievable while keeping a satisfactory quality of focusing. This study, conducted on three human skull samples, demonstrates that mechanical tilting of the transducer with the lens restores a single -3dB focal volume at 914kHz for a steering up to 11mm in the transverse direction, and 10mm in the longitudinal direction, around the focal region. This result is a step forward in using the lens-based technology in clinical applications.

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